

Borivoje Nikolic

professor

基本信息

姓名: Borivoje Nikolic

所属院校: 加州大学-伯克利

地区: United States

年龄范围: 50岁以上

企业经历: 有

学科领域

集成电路设计与集成系统

微电子科学与工程

通信工程

电子与计算机工程

研究领域

数字集成电路设计

低功耗电路与能量管理

计算机体系结构

微电子器件与制造

存储器设计与架构

嵌入式系统与硬件设计

移动通信与5G/6G

无线通信与智能天线

超大规模集成电路设计

纳米电子与先进工艺

+5

个人简介

Borivoje Nikolic is a Professor at the University of California, Berkeley, specializing in integrated circuits, computer architecture, and VLSI design. He received his Ph.D. from UC Davis in 1999 and has held positions at Silicon Systems, Inc., and Texas Instruments. His research focuses on digital and analog integrated circuit design and VLSI implementation of communication algorithms. He is co-author of "Digital Integrated Circuits: A Design Perspective" and has received numerous awards, including the NSF CAREER award and multiple best paper awards. Nikolic is also a Distinguished Lecturer of the IEEE Solid-State Circuits Society and has chaired several prestigious conferences.

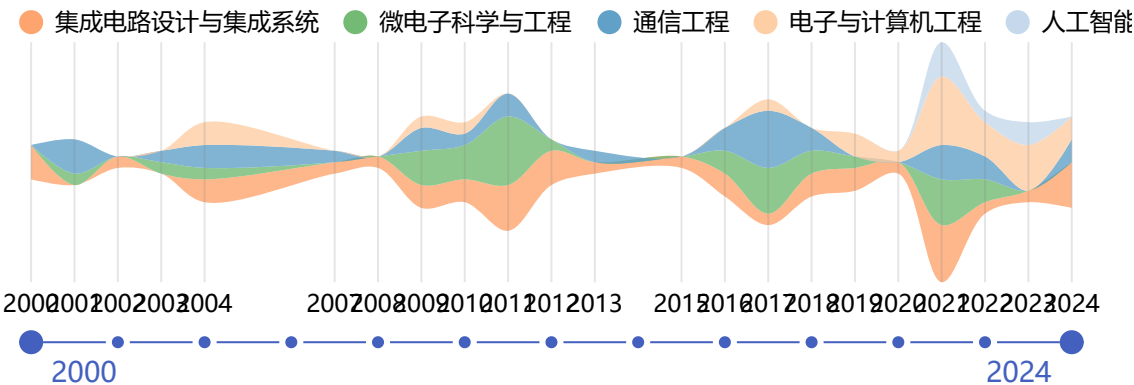
工作经历

University of Belgrade - Lecturer	1992 - 1996
Silicon Systems, Inc., Texas Instruments Storage Products Group	-
Department of Electrical Engineering and Computer Sciences, University of California at Berkeley - Professor	1999 - 至今
IEEE Solid-State Circuits Society - Distinguished Lecturer	2014 - 2015
Symposium on VLSI Technology and Circuits - Technical Program Chair	2022 - 2022
IEEE Symposium on VLSI Technology and Circuits - General Chair	2024 - 2024

教育经历

UC Davis - PhD	- 1999
University of Belgrade - MSc	- 1994
University of Belgrade - Dipl.Ing.	- 1992

近年发表的论文



FireAxe: Partitioned FPGA-Accelerated Simulation of Large-Scale RTL Designs
2024 ACM/IEEE 51ST ANNUAL INTERNATIONAL SYMPOSIUM ON COMPUTER ARCHITECTURE, ISCA 2024
Whangbo, Joonho; Lim, Edwin; Zhang, Chengyi Lux; Anderson, Kevin; Gonzalez, Abraham; Gupta, Raghav;
Krishnakumar, Nivedha; Karandikar, Sagar; Nikolic, Borivoje; Shao, Yakun Sophia; Asanovic, Krste
被引用次数: 0 2024

A Generated 4 GS/s 124.6 mW 8x Time-Interleaved SAR-VCO ADC with 9.1 ENOB
2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024
Liu, Zhaokai; Nikolic, Borivoje

被引用次数: 0

2024

A Multi-Phase Injection-Locked 8-Phase 17 GHz Clock Generator with 6b Phase Rotation for Multi-Phase Wireline Receivers

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Zhou, Bob; Nikolic, Borivoje

被引用次数: 0

2024

Design Approach for Die-to-Die Interfaces to Enable Energy-Efficient Chiplet Systems

PROCEEDINGS OF THE 29TH ACM/IEEE INTERNATIONAL SYMPOSIUM ON LOW POWER ELECTRONICS AND DESIGN, ISLPED 2024

Jain, Vikram; Tang, Wei; Wu, Zuoguo; Schmulbach, Viansa; Shao, Sophia; Zhang, Zhengya; Nikolic, Borivoje

被引用次数: 0

2024

A Chisel Generator for Standardized 3-D Die-to-Die Interconnects

IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS

Liew, Harrison; Sheikh, Farhana; Guo, Jong-Ru; Wu, Zuoguo; Nikolic, Borivoje

被引用次数: 0

2024

AuRORA: A Full-Stack Solution for Scalable and Virtualized Accelerator Integration

IEEE MICRO

Kim, Seah; Zhao, Jerry; Asanovic, Krste; Nikolic, Borivoje; Shao, Yakun Sophia

被引用次数: 0

2024

SC_μM-V23: Towards A Crystal-Free System-On-Chip For IoT In 16nm

2024 IEEE WORKSHOP ON CRYSTAL-FREE/-LESS RADIO AND SYSTEM-BASED RESEARCH FOR IOT, CRYSTALFREEIOT 2024

Lovell, Daniel; Yuan, Titan; Lin, Yu-Chi; Bapat, Maithili; Hijab, Rami; Ramachandran, Neelesh; Krause, Elisa; Patle, Nagesh; Chung, Stephen; Lee, Yun-Chieh; He, Yuqi; Khodkumbhe, Awani; Tran, Matthew; Wang, Jero; Endraws, Daniel; Jin, Qitong; Lu, Kevin; Aeron, Shrey; Le, Dang; Fawal, Tarik; Gonzalez-Serrano, Rigoberto; Nikolic, Borivoje; Niknejad, Ali M.; Pister, Kristofer S. J.

被引用次数: 0

2024

AuRORA: Virtualized Accelerator Orchestration for Multi-Tenant Workloads

IEEE/ACM International Symposium on Microarchitecture

Seah Kim; Jerry Zhao; Krste Asanovic; Borivoje Nikolic; Yakun Sophia Shao

被引用次数: 0

2023

Guest Editorial Introduction to the Special Issue on the 2022 Symposium on VLSI Circuits

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Nikolic, Borivoje; Hamada, Mototsugu

被引用次数: 0

2023

RoSE: A Hardware-Software Co-Simulation Infrastructure Enabling Pre-Silicon Full-Stack Robotics SoC Evaluation

PROCEEDINGS OF THE 2023 THE 50TH ANNUAL INTERNATIONAL SYMPOSIUM ON COMPUTER ARCHITECTURE, ISCA 2023

Nikiforov, Dima; Dong, Shengjun Kris; Zhang, Chengyi Lux; Kim, Seah; Nikolic, Borivoje; Shao, Yakun Sophia

被引用次数: 5

2023

CDPU: Co-designing Compression and Decompression Processing Units for Hyperscale Systems

PROCEEDINGS OF THE 2023 THE 50TH ANNUAL INTERNATIONAL SYMPOSIUM ON COMPUTER ARCHITECTURE, ISCA 2023

Karandikar, Sagar; Udipi, Aniruddha N.; Choi, Junsun; Whangbo, Joonho; Zhao, Jerry; Kanev, Svilen; Lim, Edwin; Alakuijala, Jyrki; Madduri, Vrishab; Shao, Yakun Sophia; Nikolic, Borivoje; Asanovic, Krste; Ranganathan, Parthasarathy

被引用次数: 3

2023

AuRORA: Virtualized Accelerator Orchestration for Multi-Tenant Workloads

56TH IEEE/ACM INTERNATIONAL SYMPOSIUM ON MICROARCHITECTURE, MICRO 2023

Kim, Seah; Zhao, Jerry; Asanovic, Krste; Nikolic, Borivoje; Shao, Yakun Sophia

被引用次数: 4

2023

Design Techniques for a 6.4-32-Gb/s 0.96-pJ/b Continuous-Rate CDR With Stochastic Frequency-Phase Detector

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Kwanseo; Shim, Minkyu; Ko, Han-Gon; Nikolic, Borivoje; Jeong, Deog-Kyoon

被引用次数: 12

2022

An Eight-Core 1.44-GHz RISC-V Vector Processor in 16-nm FinFET

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Schmidt, Colin; Wright, John; Wang, Zhongkai; Chang, Eric; Ou, Albert; Bae, Woorham; Huang, Sean; Milovanovic, Vladimir; Flynn, Anita; Richards, Brian; Asanovic, Krste; Alon, Elad; Nikolic, Borivoje

被引用次数: 7

2022

Introduction to the Special Issue on the 2021 Symposium on VLSI Circuits

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Oike, Yusuke; Nikolic, Borivoje

被引用次数: 0

2022

An Adaptable and Scalable Generator of Distributed Massive MIMO Baseband Processing Systems

JOURNAL OF SIGNAL PROCESSING SYSTEMS FOR SIGNAL IMAGE AND VIDEO TECHNOLOGY

Dai, Yue; Rasekh, Maryam Eslami; Mirfarshbafan, Seyed Hadi; Liew, Harrison; Gallyas-Sanhueza, Alexandra; Dunn, James; Madhow, Upamanyu; Studer, Christoph; Nikolic, Borivoje

被引用次数: 0

2022

Invited: Hammer: A Modular and Reusable Physical Design Flow Tool

PROCEEDINGS OF THE 59TH ACM/IEEE DESIGN AUTOMATION CONFERENCE, DAC 2022

Liew, Harrison; Grubb, Daniel; Wright, John; Schmidt, Colin; Krzysztofowicz, Nayiri; Izraelevitz, Adam; Wang, Edward; Asanovic, Krste; Bachrach, Jonathan; Nikolic, Borivoje

被引用次数: 10

2022

Automated Design of Analog Circuits Using Reinforcement Learning

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Settaluri, Keertana; Liu, Zhaokai; Khurana, Rishabh; Mirhaj, Arash; Jain, Rajeev; Nikolic, Borivoje

被引用次数: 17

2022

Constellation: An Open-Source SoC-Capable NoC Generator

2022 15TH IEEE/ACM INTERNATIONAL WORKSHOP ON NETWORK ON CHIP ARCHITECTURES (NOCARC 2022)

Zhao, Jerry; Agrawal, Animesh; Nikolic, Borivoje; Asanovic, Krste

被引用次数: 16

2022

An Automated and Process-Portable Generator for Phase-Locked Loop

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Wang, Zhongkai; Choi, Minsoo; Chang, Eric; Wright, John; Bae, Wooham; Du, Sijun; Liu, Zhaokai; Narevsky, Nathan; Schmidt, Colin; Biwas, Ayan; Nikolic, Borivoje; Alon, Elad

被引用次数: 2

2021

Gemmini: Enabling Systematic Deep-Learning Architecture Evaluation via Full-Stack Integration

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Genc, Hasan; Kim, Seah; Amid, Alon; Haj-Ali, Ameer; Iyer, Vighnesh; Prakash, Pranav; Zhao, Jerry; Grubb, Daniel; Liew, Harrison; Mao, Howard; Ou, Albert; Schmidt, Colin; Steffl, Samuel; Wright, John; Stoica, Ion; Ragan-Kelley, Jonathan; Asanovic, Krste; Nikolic, Borivoje; Shao, Yakun Sophia

被引用次数: 110

2021

A Scalable Massive MIMO Uplink Baseband Processing Generator

IEEE INTERNATIONAL CONFERENCE ON COMMUNICATIONS (ICC 2021)

Dai, Yue; LaCaille, Greg; Liew, Harrison; Dunn, James; Nikolic, Borivoje

被引用次数: 4

2021

LAYGO: A Template-and-Grid-Based Layout Generation Engine for Advanced CMOS Technologies

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Han, Jaeduk; Bae, Wooham; Chang, Eric; Wang, Zhongkai; Nikolic, Borivoje; Alon, Elad

被引用次数: 19

2021

Gemmini: Enabling Systematic Deep-Learning Architecture Evaluation via Full-Stack Integration

Design Automation Conference

Hasan Genc; Seah Kim; Alon Amid; Ameer Haj-Ali; Vighnesh Iyer; Pranav Prakash; Jerry Zhao; Daniel Grubb; Harrison Liew; Howard Mao; Albert Ou; Colin Schmidt; Samuel Steffl; John Wright; Ion Stoica; Jonathan Ragan-Kelley; Krste Asanovic; Borivoje Nikolic; Yakun Sophia Shao

被引用次数: 0

2021

An Automated and Process-Portable Generator for Phase-Locked Loop

Design Automation Conference

Zhongkai Wang; Minsoo Choi; Eric Chang; John Wright; Wooham Bae; Sijun Du; Zhaokai Liu; Nathan Narevsky; Colin Schmidt; Ayan Biwas; Borivoje Nikolic; Elad Alon

被引用次数: 0

2021

A Hardware Accelerator for Protocol Buffers

IEEE/ACM International Symposium on Microarchitecture

Sagar Karandikar; Chris Leary; Chris Kennelly; Jerry Zhao; Dinesh Parimi; Borivoje Nikolic; Krste Asanovic; Parthasarathy Ranganathan

被引用次数: 0

2021

Improved Processor Design Project Testing

2021 ACM/IEEE WORKSHOP ON COMPUTER ARCHITECTURE EDUCATION (WCAE)

Lund, Ryan; McMahon, Connor; Garcia, Dan; Nikolic, Borivoje

被引用次数: 1

2021

A Scalable Generator for Massive MIMO Baseband Processing Systems with Beam-space Channel Estimation

2021 IEEE WORKSHOP ON SIGNAL PROCESSING SYSTEMS (SIPS 2021)

Dai, Yue; Liew, Harrison; Rasekh, Maryam Eslami; Mirfarshbafan, Seyed Hadi; Gallyas-Sanhueza, Alexandra; Dunn, James; Madhow, Upamanyu; Studer, Christoph; Nikolic, Borivoje

被引用次数: 1

2021

A 16mm² 106.1 GOPS/W Heterogeneous RISC-V Multi-Core Multi-Accelerator SoC in Low-Power 22nm FinFET

ESSCIRC 2021 - IEEE 47TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE (ESSCIRC)

Gonzalez, Abraham; Zhao, Jerry; Korpan, Ben; Genc, Hasan; Schmidt, Colin; Wright, John; Biswas, Ayan; Amid, Alon; Sheikh, Farhana; Sorokin, Anton; Kale, Sirisha; Yalamanchi, Mani; Yarlagadda, Ramya; Flannigan, Mark; Abramowitz, Larry; Alon, Elad; Shao, Yakun Sophia; Asanovic, Krste; Nikolic, Borivoje

被引用次数: 13

2021

Accelerator Integration for Open-Source SoC Design

IEEE MICRO

Biancolin, David; Magyar, Albert; Karandikar, Sagar; Amid, Alon; Nikolic, Borivoje; Bachrach, Jonathan; Asanovic, Krste

被引用次数: 4

2021

A 71-to-86-GHz 16-Element by 16-Beam Multi-User Beamforming Integrated Receiver Sub-Array for Massive MIMO

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Naviasky, Emily; Iotti, Lorenzo; LaCaille, Greg; Nikolic, Borivoje; Alon, Elad; Niknejad, Ali M.

被引用次数: 19

2021

A Line-Array Technique for Wireless Power Transfer Toward a 100 μm x 100 μm Coil Antenna

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Zhao, Bo; Kuo, Nai-Chung; Niknejad, Ali M.; Nikolic, Borivoje

被引用次数: 4

2020

A Dual-Core RISC-V Vector Processor With On-Chip Fine-Grain Power Management in 28-nm FD-SOI

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Wright, John Charles; Schmidt, Colin; Ben Keller; Dabbelt, Daniel Palmer; Kwak, Jaehwa; Iyer, Vighnesh; Mehta, Nandish; Chiu, Pi-Feng; Bailey, Stevo; Asanovic, Krste; Nikolic, Borivoje

被引用次数: 9

2020

Chipyard: Integrated Design, Simulation, and Implementation Framework for Custom SoCs

IEEE MICRO

Amid, Alon; Biancolin, David; Gonzalez, Abraham; Grubb, Daniel; Karandikar, Sagar; Liew, Harrison; Magyar, Albert; Mao, Howard; Ou, Albert; Pemberton, Nathan; Rigge, Paul; Schmidt, Colin; Wright, John; Zhao, Jerry; Shao, Yakun Sophia; Asanovic, Krste; Nikolic, Borivoje

被引用次数: 162

2020

FireSim: FPGA-Accelerated Cycle-Exact Scale-Out System Simulation in the Public Cloud

IEEE MICRO

Karandikar, Sagar; Mao, Howard; Kim, Donggyu; Biancolin, David; Amid, Alon; Lee, Dayeol; Pemberton, Nathan; Amaro, Emmanuel; Schmidt, Colin; Chopra, Aditya; Huang, Qijing; Kovacs, Kyle; Nikolic, Borivoje; Katz, Randy Howard; Bachrach, Jonathan; Asanovic, Krste

被引用次数: 6

2019

BROOM: An Open Source Out-of-Order Processor With Resilient Low-Voltage Operation in 28-nm CMOS

IEEE MICRO

Cello, Christopher; Chin, Pi-Feng; Asanovic, Krste; Nikolic, Borivoje; Patterson, David

被引用次数: 9

2019

A Mixed-Signal RISC-V Signal Analysis SoC Generator With a 16-nm FinFET Instance

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bailey, Steven; Rigge, Paul; Han, Jaeduk; Lin, Richard; Chang, Eric Y.; Mao, Howard; Wang, Zhongkai; Markley, Chick; Izraelevitz, Adam M.; Wang, Angie; Narevsky, Nathan; Bae, Woorham; Shauck, Steve; Montano, Sergio; Norsworthy, Justin; Razzaque, Munir; Ma, Wen Hau; Lentiro, Akalu; Doerflein, Matthew; Heckendorn, Darin; McGrath, Jim; DeSeta, Franco; Shoham, Ronen; Stellfox, Mike; Snowden, Mark; Cole, Joseph; Fuhrman, Daniel R.; Richards, Brian; Bachrach, Jonathan; Alon, Elad; Nikolic, Borivoje

被引用次数: 11

2019

A Real-Time, 1.89-GHz Bandwidth, 175-kHz Resolution Sparse Spectral Analysis RISC-V SoC in 16-nm FinFET

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Wang, Angie; Bae, Woorham; Han, Jaeduk; Bailey, Stevo; Ocal, Orhan; Rigge, Paul; Wang, Zhongkai; Ramchandran, Kannan; Alon, Elad; Nikolic, Borivoje

被引用次数: 7

2019

A Variation-Tolerant, Sneak-Current-Compensated Readout Scheme for Cross-Point Memory Based on Two-Port Sensing Technique

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Bae, Woorham; Yoon, Kyung Jean; Song, Taeksang; Nikolic, Borivoje

A 0.4-to-4-GHz All-Digital RF Transmitter Package With a Band-Selecting Interposer Combining Three Wideband CMOS Transmitters

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Kuo, Nai-Chung; Yang, Bonjern; Wang, Angie; Kong, Lingkai; Wu, Charles; Srin, Vason P; Alon, Elad; Nikolic, Borivoje; Niknejad, Ali M.

被引用次数: 11

2018

A 65-nm CMOS I / Q RF Power DAC With 24-to 42-dB Third-Harmonic Cancellation and Up to 18-dB Mixed-Signal Filtering

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yang, Bonjern; Chang, Eric Y.; Niknejad, Ali M.; Nikolic, Borivoje; Alon, Elad

被引用次数: 16

2018

Reprogrammable Redundancy for SRAM-Based Cache Vmin Reduction in a 28-nm RISC-V Processor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zimmer, Brian; Chiu, Pi-Feng; Nikolic, Borivoje; Asanovic, Krste

被引用次数: 5

2017

Analysis and Design of Integrated Active Cancellation Transceiver for Frequency Division Duplex Systems

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Calderin, Lucas; Ramakrishnan, Sameet; Puglielli, Antonio; Alon, Elad; Nikolic, Borivoje; Niknejad, Ali M.

被引用次数: 41

2017

Use of Phase Delay Analysis for Evaluating Wideband Circuits: An Alternative to Group Delay Analysis

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Bae, Woorham; Nikolic, Borivoje; Jeong, Deog-Kyoon

被引用次数: 8

2017

Real-Time Cooperative Communication for Automation Over Wireless

IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS

Swamy, Vasuki Narasimha; Suri, Sahaana; Rigge, Paul; Weiner, Matthew; Ranade, Gireeja; Sahai, Anant; Nikolic, Borivoje

被引用次数: 38

2017

A 65-nm CMOS Wideband TDD Front-End With Integrated T/R Switching via PA Re-Use

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Xiao, Xiao; Pratt, Amanda; Yang, Bonjern; Wang, Angie; Niknejad, Ali M.; Alon, Elad; Nikolic, Borivoje

被引用次数: 12

2017

A RISC-V Processor SoC With Integrated Power Management at Submicrosecond Timescales in 28 nm FD-SOI

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Keller, Ben; Cochet, Martin; Zimmer, Brian; Kwak, Jaehwa; Puggelli, Alberto; Lee, Yunsup; Blagojevic, Milovan; Bailey, Stevo; Chiu, Pi-Feng; Dabbelt, Palmer; Schmidt, Colin; Alon, Elad; Asanovic, Krste; Nikolic, Borivoje

被引用次数: 30

2017

A Wideband All-Digital CMOS RF Transmitter on HDI Interposers With High Power and Efficiency

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Kuo, Nai-Chung; Yang, Bonjern; Wang, Angie; Kong, Ling kai; Wu, Charles; Srin, Vason P; Alon, Elad; Nikolic, Borivoje; Niknejad, Ali M.

被引用次数: 18

2017

AN AGILE APPROACH TO BUILDING RISC-V MICROPROCESSORS

IEEE MICRO

Lee, Yunsup; Waterman, Andrew; Cook, Henry; Zimmer, Brian; Keller, Ben; Puggelli, Alberto; Kwak, Jaehwa; Jevtic, Ruzica; Bailey, Stevo; Blagojevic, Milovan; Chiu, Pi-Feng; Avizienis, Rimas; Richards, Brian; Bachrach, Jonathan; Patterson, David; Alon, Elad; Nikolic, Borivoje; Asanovic, Krste

被引用次数: 80

2016

Design of Energy-and Cost-Efficient Massive MIMO Arrays

PROCEEDINGS OF THE IEEE

Puglielli, Antonio; Townley, Andrew; LaCaille, Greg; Milovanovic, Vladimir; Lu, Pengpeng; Trotskovsky, Konstantin; Whitcombe, Amy; Narevsky, Nathan; Wright, Gregory; Courtade, Thomas; Alon, Elad; Nikolic, Borivoje; Niknejad, Ali M.

被引用次数: 121

2016

A Self-Adjustable Clock Generator With Wide Dynamic Range in 28 nm FDSOI

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kwak, Jaehwa; Nikolic, Borivoje

被引用次数: 12

2016

A RISC-V Vector Processor With Simultaneous-Switching Switched-Capacitor DC-DC Converters in 28 nm FDSOI

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zimmer, Brian; Lee, Yunsup; Puggelli, Alberto; Kwak, Jaehwa; Jevtic, Ruzica; Keller, Ben; Bailey, Steven; Blagojevic, Milovan; Chiu, Pi-Feng; Hanh-Phuc Le; Chen, Po-Hung; Sutardja, Nicholas; Avizienis, Rimas; Waterman, Andrew; Richards, Brian; Flatresse, Philippe; Alon, Elad; Asanovic, Krste; Nikolic, Borivoje

被引用次数: 49

2016

An Interference-Resilient Wideband Mixer-First Receiver With LO Leakage Suppression and I/Q Correlated Orthogonal Calibration

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Wu, Charles; Wang, Yanjie; Nikolic, Borivoje; Hull, Christopher

被引用次数: 36

2016

A Differential 2R Crosspoint RRAM Array With Zero Standby Current

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chiu, Pi-Feng; Nikolic, Borivoje

Per-Core DVFS With Switched-Capacitor Converters for Energy Efficiency in Manycore Processors

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Jevtic, Ruzica; Hanh-Phuc Le; Blagojevic, Milovan; Bailey, Stevo; Asanovic, Krste; Alon, Elad; Nikolic, Borivoje

被引用次数: 36

2015

Coding and System Design for Quantize-Map-and-Forward Relaying

IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS

Nagpal, Vinayak; Wang, I-Hsiang; Jorgovanovic, Milos; Tse, David; Nikolic, Borivoje

被引用次数: 10

2013

A 2.8 GS/s 44.6 mW Time-Interleaved ADC Achieving 50.9 dB SNDR and 3 dB Effective Resolution Bandwidth of 1.5 GHz in 65 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Stepanovic, Dusan; Nikolic, Borivoje

被引用次数: 177

2013

SRAM Assist Techniques for Operation in a Wide Voltage Range in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zimmer, Brian; Toh, Seng Oon; Vo, Huy; Lee, Yunsup; Thomas, Olivier; Asanovic, Krste; Nikolic, Borivoje

被引用次数: 60

2012

Multiple-Input Relay Design for More Compact Implementation of Digital Logic Circuits

IEEE ELECTRON DEVICE LETTERS

Jeon, Jaeseok; Hutin, Louis; Jevtic, Ruzica; Liu, Nathaniel; Chen, Yenhao; Nathanael, Rhessa; Kwon, Wookhyun; Spencer, Matthew; Alon, Elad; Nikolic, Borivoje; Liu, Tsu-Jae King

被引用次数: 18

2012

A 15 MHz to 600 MHz, 20 mW, 0.38 mm² Split-Control, Fast Coarse Locking Digital DLL in 0.13 μ m CMOS

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Hoyos, Sebastian; Tsang, Cheongyuen W.; Vanderhaegen, Johan; Chiu, Yun; Aibara, Yasutoshi; Khorramabadi, Haideh; Nikolic, Borivoje

被引用次数: 18

2012

Discrete-Time, Linear Periodically Time-Variant Phase-Locked Loop Model for Jitter Analysis

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Vamvakos, Socrates D.; Stojanovic, Vladimir; Nikolic, Borivoje

被引用次数: 13

2011

A 2 Gb/s 5.6 mW Digital LOS/NLOS Equalizer for the 60 GHz Band

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Ji-Hoon; Richards, Brian; Nikolic, Borivoje

被引用次数: 12

2011

Characterization of Dynamic SRAM Stability in 45 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Toh, Seng Oon; Guo, Zheng; Liu, Tsu-Jae King; Nikolic, Borivoje

被引用次数: 50

2011

Quasi-planar bulk CMOS technology for improved SRAM scalability

SOLID-STATE ELECTRONICS

Shin, Changhwan; Tsai, Chen Hua; Wu, Mei Hsuan; Chang, Chung Fu; Liu, You Ren; Kao, Chih Yang; Lin, Guan Shyan; Chiu, Kai Ling; Fu, Chuan-Shian; Tsai, Cheng-tzung; Liang, Chia Wen; Nikolic, Borivoje; Liu, Tsu-Jae King

被引用次数: 8

2011

Technology Variability From a Design Perspective

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Nikolic, Borivoje; Park, Ji-Hoon; Kwak, Jaehwa; Giraud, Bastien; Guo, Zheng; Pang, Liang-Teck; Toh, Seng Oon; Jevtic, Ruzica; Qian, Kun; Spanos, Costas

被引用次数: 17

2011

Performance and Yield Benefits of Quasi-Planar Bulk CMOS Technology for 6-T SRAM at the 22-nm Node

IEEE TRANSACTIONS ON ELECTRON DEVICES

Shin, Changhwan; Damrongplasit, Nattapol; Sun, Xin; Tsukamoto, Yasumasa; Nikolic, Borivoje; Liu, Tsu-Jae King

被引用次数: 22

2011

Performance and Area Scaling Benefits of FD-SOI Technology for 6-T SRAM Cells at the 22-nm Node

IEEE TRANSACTIONS ON ELECTRON DEVICES

Shin, Changhwan; Cho, Min Hee; Tsukamoto, Yasumasa; Nguyen, Bich-Yen; Mazure, Carlos; Nikolic, Borivoje; Liu, Tsu-Jae King

被引用次数: 41

2010

Analysis of Absorbing Sets and Fully Absorbing Sets of Array-Based LDPC Codes

IEEE TRANSACTIONS ON INFORMATION THEORY

Dolecek, Lara; Zhang, Zhengya; Anantharam, Venkat; Wainwright, Martin J.; Nikolic, Borivoje

被引用次数: 156

2010

SRAM Read/Write Margin Enhancements Using FinFETs

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